

7.3 Test Cases for Website functioning and Question Recommendation

Test Case 1

- **Test Case :** User Registration with Valid Credentials
 - **Test Statement:** Verify that a user can successfully register using valid credentials.
 - **Steps:**
 1. Open the website.
 2. Click on the "Sign Up" button.
 3. Enter a valid name, email, and password.
 4. Click on "Register."
 - **Expected Result:** User is successfully registered and redirected to the dashboard.
 - **Pass/Fail:** Pass
-

Test Case 2:

- **Test Case :** User Registration with Invalid Email
 - **Test Statement:** Ensure that the system validates incorrect email formats.
 - **Steps:**
 1. Enter an invalid email format (e.g., "abc@xyz").
 2. Click on "Register."
 - **Expected Result:** The system displays a validation error: "Invalid email format."
 - **Pass/Fail:** Pass
-

Test Case 3:

- **Test Case :** Login with Valid Credentials
 - **Test Statement :** Verify that a user can log in with valid credentials.
 - **Steps:**
 1. Open the login page.
 2. Enter a registered email and correct password.
 3. Click on "Login."
 - **Expected Result:** User successfully logs in and is redirected to the home page.
 - **Pass/Fail: Pass**
-

Test Case 4:

- **Test Case :** Login with Invalid Password
 - **Test Statement:** Ensure that incorrect password login attempts are handled properly.
 - **Steps:**
 - a. Enter a valid email but an incorrect password.
 - b. Click on "Login."
 - **Expected Result:** The system displays an error message: "Invalid credentials."
 - **Pass/Fail: Pass**
-

Test Case 5:

- **Test Case :** Auto-Login with Cookies
- **Test Statement:** Verify that the system remembers logged-in users using cookies.
- **Steps:**
 1. Log in successfully.

2. Close and reopen the browser.
 3. Open the website again.
- **Expected Result:** The user remains logged in if cookies are enabled.
 - **Pass/Fail:** Pass
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Test Case 6:

- **Test Case :** Logout Functionality
 - **Test Statement:** Verify that users can log out and sessions are cleared.
 - **Steps:**
 1. Click on the "Logout" button.
 - **Expected Result:** The user is redirected to the login page, and their session is cleared.
 - **Pass/Fail:** Pass
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Test Case 7:

- **Test Case : Recommendation System - User Profile Based**
 - **Test Statement:** Verify that recommendations are personalized based on previous attempts.
 - **Steps:**
 1. Solve five or more questions on a specific topic.
 2. Refresh the page.
 - **Expected Result:** New recommended questions should be tailored based on past performance.
 - **Pass/Fail:** Pass
-

Test Case 8:

- **Test Case :**Recommendation System - Sequential Learning
 - **Test Statement:** Ensure that question difficulty increases as the user progresses.
 - **Steps:**
 1. Answer a series of easy questions correctly.
 2. Check the next recommended question.
 - **Expected Result:** The system gradually increases question difficulty.
 - **Pass/Fail:** Pass
-

Test Case 9:

- **Test Case :** API Response Time for Recommendations
 - **Test Statement:** Verify that recommendation API responds within an acceptable time.
 - **Steps:**
 1. Open the dashboard.
 2. Inspect API response time using Developer Tools (Network tab).
 - **Expected Result:** The API should respond within 2 seconds.
 - **Pass/Fail:** Fail
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Test Case 10:

- **Test Case :** Handling of Incorrect API Response
- **Test Statement:** Ensure that system handles API failures gracefully.
- **Steps:**
 1. Simulate an API failure (e.g., disconnect the internet).
 2. Try fetching recommendations.
- **Expected Result:** The system should show an appropriate error message like "Unable to fetch recommendations. Please try again."

- **Pass/Fail: Fail**
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Test Case 11:

- **Test Case :** Progress Tracking Visualization
 - **Test Statement:** Verify that user progress tracking updates correctly.
 - **Steps:**
 1. Solve multiple questions.
 2. Navigate to the "Progress Tracking" page.
 - **Expected Result:** User's performance chart updates accurately.
 - **Pass/Fail: Pass**
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Test Case 12:

- **Test Case :** UI Responsiveness - Mobile View
 - **Test Statement:** Ensure that the UI is properly displayed on mobile devices.
 - **Steps:**
 1. Open the website on a mobile device.
 2. Check UI elements (buttons, text fields, responsiveness).
 - **Expected Result:** The layout adjusts properly without overlapping elements.
 - **Pass/Fail: Fail**
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Test Case 13:

- **UI Responsiveness - Desktop View**
- **Test Statement:** Verify that the UI adapts properly to different desktop resolutions.
- **Steps:**
 1. Open the website on a desktop.
 2. Resize the browser window.

- **Expected Result:** The layout should adjust dynamically without breaking.
 - **Pass/Fail:** Pass
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Test Case 14:

- **Test Case :** Security - SQL Injection Prevention
 - **Test Statement:** Ensure that the system is protected against SQL Injection attacks.
 - **Steps:**
 1. Enter " ' OR 1=1 --" in any input field.
 2. Click Submit.
 - **Expected Result:** The system sanitizes the input and prevents SQL injection.
 - **Pass/Fail:** Fail
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Test Case 15:

- **Test Case :** Security - JWT Token Expiry
- **Test Statement:** Verify that session expiration is handled correctly.
- **Steps:**
 1. Log in successfully.
 2. Wait for the JWT token to expire.
 3. Try accessing a protected page.
- **Expected Result:** The user is redirected to the login page after session expiry.
- **Pass/Fail:** Pass

7.4 Testing for Adaptive Learning in Question Recommendation

Test Case - 1

- **Test Case:** Increase question difficulty after high acceptance rate
 - **Test Statement:** Verify that the system increases question difficulty when the user consistently answers correctly.
 - **Steps:**
 - a. Answer 5 or more questions of similar difficulty correctly.
 - b. Refresh the page or check the next recommended question.
 - **Expected Result:** The system should recommend a harder question.
 - **Actual Result:** Question is harder but not as it should.
 - **Feedback Source:** This was observed by users and flagged by the JEE coaching mentor.
 - **Pass/Fail:** *Fail*
-

Test Case - 2

- **Test Case:** Decrease difficulty after consecutive failures
 - **Test Statement:** Verify that the system reduces question difficulty when the user repeatedly answers incorrectly.
 - **Steps:**
 - a. Answer 3 questions incorrectly in a row.
 - b. Refresh the page or check the next recommended question.
 - **Expected Result:** The system does not lower the difficulty level as expected. Even after three consecutive incorrect answers, the next question remained at the same or a slightly higher difficulty.
 - **Feedback Source:** This was observed by internal testers and flagged by the JEE coaching mentor.
 - **Actual Result:** It is not working fine,
 - **Pass/Fail:** *Fail*
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Test Case - 3

- **Test Case:** Maintain difficulty level for mixed performance
 - **Test Statement:** Verify that the system does not change difficulty drastically when user performance is inconsistent.
 - **Steps:**
 - a. Answer some questions correctly and some incorrectly.
 - b. Check the next question.
 - **Expected Result:** The system should maintain similar difficulty until a clear trend is identified.
 - **Actual Result:** The system maintains a steady difficulty level, showing only slight variation, which is appropriate for consistent user performance.
 - **Pass/Fail:** Pass
-

7.5 Test Cases for Future Testing Purposes (To Be Tested when Sufficient User Data is Available)

These test cases include scenarios that cannot be effectively tested at present due to limited user activity or insufficient data. As the platform gains more users and interactions, these cases will become testable to ensure scalability, accuracy, and adaptive behavior in a real-world environment.

Test Case - 4

- **Test Case:** Adapt difficulty based on time taken (fast response)
- **Test Statement:** Verify that the system increases question difficulty if the user answers significantly faster than the average time.
- **Steps:**
 - a. Solve 5+ questions under the average time taken by most users.
 - b. Check the next recommended question.
- **Expected Result:** The system should recommend a harder question.
- **Actual Result:** *To be tested*

- **Pass/Fail:** *To be recorded*
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Test Case - 5

- **Test Case:** Adapt difficulty based on time taken (slow response)
 - **Test Statement:** Verify that the system decreases question difficulty if the user takes too long to answer.
 - **Steps:**
 - a. Take longer than the average time to answer 3+ questions.
 - b. Check the next recommended question.
 - **Expected Result:** The system should recommend an easier question.
 - **Actual Result:** *To be tested*
 - **Pass/Fail:** *To be recorded*
-

Test Case - 6

- **Test Case:** Edge Case - User answers randomly
 - **Test Statement:** Verify that the system does not change question difficulty abruptly when answers are random.
 - **Steps:**
 - a. Answer questions randomly (mix of correct & incorrect with no pattern).
 - b. Check if difficulty changes.
 - **Expected Result:** The system should wait for a clearer trend before adapting difficulty.
 - **Actual Result:** *To be tested*
 - **Pass/Fail:** *To be recorded*
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Test Case - 7

- **Test Case:** Verify adaptation over long sessions
 - **Test Statement:** Verify that the system gradually adjusts difficulty over an extended learning session.
 - **Steps:**
 - a. Solve 20+ questions over a long session.
 - b. Track the difficulty level changes.
 - **Expected Result:** The difficulty level should adjust gradually rather than suddenly.
 - **Actual Result:** *To be tested*
 - **Pass/Fail:** *To be recorded*
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Test Case - 8

- **Test Case:** Test difficulty adaptation for multiple users
- **Test Statement:** Verify that different users with different performances get different difficulty levels.
- **Steps:**
 - a. Use two different user accounts.
 - b. Compare the recommended questions for both users.
- **Expected Result:** The system should recommend different difficulty levels based on each user's performance.
- **Actual Result:** *To be tested*
- **Pass/Fail:** *To be recorded*

7.5 Testing for Question Recommendation

Test Case - 1

- **Test Case:** Test recommendation of question by LSTM model

No	Test Case	User Level	Expected Result	Actual Result	Pass/Fail
1	$\int(1+x+x^2/2!+x^3/3!+\dots) dx=$	1	$\int(a x^{-2} + b x^{-1} + c)/x^{-3} dx=$	If $1/((1+x)\sqrt{x}) dx = f(x) + A$	FAIL
2	$\int(1+x+x^2/2!+x^3/3!+\dots) dx=$	5	If $\int(\cos 2x - \cos 2\alpha)/(\cos x - \cos \alpha) d\alpha = f(\alpha) + A\alpha$	If $\int(\cos 2x - \cos 2\alpha)/(\cos x - \cos \alpha) d\alpha = f(\alpha) + A\alpha$	PASS
3	$\int(1+x+x^2/2!+x^3/3!+\dots) dx=$	10	$\int e^x (\cos x - \sin x) dx$	$\int e^x (\cos x - \sin x) dx$	PASS
4	If $f'(x) = 1/x + x$ and $f(1) = 5/2$, then $f(x) =$	1	$\int(1 - x^7)/(x(1 + x^7)) dx =$	If $f(x) = g(x)$, then $\int f'(x)g(x) dx =$	FAIL
5	$\int dx/((x+2)(x^2+1)) = a \ln(1+x^2) + b \tan^{-1}x + 1/5 \ln x+2 + C$	5	$\int dx/((x+2)(x^2+1)) = a \ln(1+x^2) + b \tan^{-1}x + 1/5 \ln x+2 + C$	$\int dx/((x+2)(x^2+1)) = a \ln(1+x^2) + b \tan^{-1}x + 1/5 \ln x+2 + C$	PASS

6	If $f(x) = 1/x + x$ and $f(1) = 5/2$, then $f(x) =$	10	$\int dx/((x^2+1)(x^2+4)) =$	$\int \tan^{-1}\sqrt{(1-\cos 2x)/(1+\cos 2x))} dx =$	FAIL
7	$\int (1 - x^7)/(x(1 + x^7)) dx =$	2	If $f(x) = g(x)$, then $\int f(x)g(x) dx =$	If $f(x) = g(x)$, then $\int f(x)g(x) dx =$	PASS
8	$\int (1 - x^7)/(x(1 + x^7)) dx =$	6	$\int (x^2 + x - 6)/((x-2)(x-1)) dx =$	If $\int dx/(5 + 4\cos x) = K \tan^{-1}(M \tan(x/2)) + c$	FAIL
9	$\int (1 - x^7)/(x(1 + x^7)) dx =$	9	$\int dx/((x^2+1)(x^2+4)) =$	$\int x^2/\sqrt{1-x} dx =$	FAIL
10	$\int (1 + \cos^2 x)/\sin^2 x dx =$	3	$\int dx/(\sin^2 x \cos^2 x) =$	$\int \sqrt{x^5}/(\sqrt{x^7 + x^6}) dx = a \ln(x^k/(x^K + 1)) + C$	FAIL

Test Case - 2

- **Test Case:** Test recommendation of question by GRU model

No	Test Case	User Level	Expected Result	Actual Result	Pass/Fail
1	$\int (1+x+x^2/2!+x^3/3!+\dots) dx =$	1	$\int (a x^{-2} + b x^{-1} + c)/x^{-3} dx =$	$\int (x^3)/(x+1) dx$ is equal to	PASS
2	$\int (1+x+x^2/2!+x^3/3!+\dots) dx =$	5	If $\int (\cos 2x - \cos 2\alpha)/(\cos x - \cos \alpha) d\alpha = f(\alpha) + A\alpha$	$\int (x^2 + x - 6)/((x-2)(x-1)) dx =$	FAIL
3	$\int (1+x+x^2/2!+x^3/3!+\dots) dx =$	10	$\int e^x (\cos x - \sin x) dx$	$\int (x^3)/(x+1) dx$ is equal to	FAIL
4	If $f'(x)=1/x + x$ and $f(1)=5/2$, then $f(x) =$	1	$\int (1 - x^7)/(x(1 + x^7)) dx =$	$\int (a x^3 + b x^2 + c)/x^4 dx =$	FAIL
5	If $f'(x)=1/x + x$ and $f(1)=5/2$, then $f(x) =$	5	$\int dx/((x+2)(x^2+1)) = a \ln(1+x^2) + b \tan^{-1}x + 1/5 \ln x+2 + C$	$\int (a x^3 + b x^2 + c)/x^4 dx =$	FAIL
6	If $\int 1/(x + x^5) dx = f(x) + c$, then $\int (x^4)/(x + x^5) dx =$	2	$\int dx/((x^2+1)(x^2+4)) =$	$\int (1+x+x^2/2!+x^3/3!+\dots) dx =$	FAIL
7	$\int \sqrt{1 + 2\tan x(\tan x + \sec x)} dx =$	10	$\int (1 - x^7)/(x(1 + x^7)) dx =$	$\int (1 - x^7)/(x(1 + x^7)) dx =$	PASS

8	$\int \sqrt{1 + 2\tan x(\tan x + \sec x)} dx =$	6	$\int (x^2 + x - 6)/((x - 2)(x - 1)) dx =$	$\int 13^x dx$	FAIL
9	$\int dx/((x^2+1)(x^2+4)) =$	3	$\int dx/((x^2+1)(x^2+4)) =$	$\int dx/((x^2+1)(x^2+4)) =$	PASS
10	If $f'(x)=1/x + x$ and $f(1)=5/2$, then $f(x) =$	1	$\int dx/(\sin^2 x \cos^2 x) =$	$\int (x^3)/(x+1) dx$ is equal to	FAIL

Chapter 8

CONCLUSION

This chapter presents the conclusion of the work, drawn from the analysis and design phases. Section 8.1 outlines the Features of the proposed system, Section 8.2 explains the benefits of the proposed system, Section 8.3 describes the Limitations of the proposed approach and 8.4 future scope for its enhancement.

8.1 Features

The features of the project are aimed at providing an advanced adaptive learning system tailored specifically for JEE aspirants. The platform uses state-of-the-art algorithms like LSTMs to offer personalized recommendations based on sequential learning. Key features include:

- **Sequential Learning Algorithm:**
 - a. The use of LSTM (Long Short-Term Memory) networks to process and analyze sequential user performance data.
 - b. Questions are assigned embeddings, and the system predicts the next question based on these embeddings.
- **Adaptive Levels and Difficulty:**
 - a. Questions are categorized into difficulty levels, dynamically adjusting to a user's current skill level and progress.
 - b. User levels are graded on a scale from 1 to 10, adapting as users solve questions, making learning personalized and effective.
- **Integration of Points System:**
 - a. Points ranging from 1,000 to 10,000 are assigned to questions, motivating users to complete harder tasks while tracking their performance milestones.
- **Recommendation System:**

- a. A highly efficient recommendation engine predicts the next set of questions by analyzing previous attempts and overall user performance.

8.2 Benefits of this Project

- **Personalized Learning at One Place:**
 - a. The platform brings all the tools and resources for personalized question solving into a single system, reducing the hassle of switching between multiple apps or materials.
- **Modular Design:**
 - a. The system uses well-defined modules for managing users, questions, performance tracking, and recommendations, ensuring scalability and easy debugging.
- **Digitized Adaptive Model:**
 - a. The use of sequential algorithms for digitized adaptive learning ensures better preparation outcomes and modernizes traditional coaching methodologies.
- **Real-Time Difficulty Adjustment:**
 - a. Instant adjustments in difficulty levels provide tailored learning experiences, helping users focus on their weak areas more effectively.
- **Efficient Progress Tracking :**
 - a. The platform employs interactive visualizations using tools like Chart.js to monitor user performance, providing real-time insights into strengths, weaknesses, and overall progress.

8.3 Limitations of the Proposed Approach

- Despite its potential, the system has a few limitations:
- **Dependency on Network Connectivity:** The web app requires an active internet connection to function.
- **Limited Rights:** Currently, only administrators have rights for moderation and major changes.

- **Manual Verification:** User onboarding still involves manual verification for some features, limiting automation.

8.4 Future Enhancements

Future developments for the project could focus on:

- **Video Solutions and Live Interaction:**
 - a. Integrating features for video solutions and live chat to enhance interactivity.
 - b. Including live classes similar to coaching institutes for real-time guidance.
- **Advanced Algorithms:**
 - a. Exploring advanced models like Transformers for better understanding of question embeddings and sequential predictions.
 - b. Tweaking existing LSTM models to improve efficiency and adapt to edge cases.
- **Gamification Features:**
 - a. Adding gamification elements like achievement badges, leaderboards, and streak tracking to improve user engagement and motivation.
- **Multi-Language Support:**
 - a. Expanding the platform to support multiple languages, making it accessible to a wider audience across diverse demographics.
- **Mobile Application Development:**
 - a. Building a dedicated mobile app to improve accessibility and provide an offline mode for question-solving.

Appendix A

Appendix

A.1 Installation Guide

A.1.1 Introduction

JEECode is a web application designed with a comprehensive and modular approach, leveraging the MERN stack and advanced machine learning techniques to ensure a seamless and adaptive JEE preparation experience. This Guide provides step-by-step instructions for installing the application on your local machines.

A.1.2 Pre Requisite

Before installing the JEECode, ensure that you have met the following requirements:

- 1.ReactJS Version 18.2.0
- 2.VS Code Version 1.80.0 or later
- 3.JDK 17 or later
- 4.Node.js Version: v18.17.1 or later
- 5.npm Version: v9.6.7 or later (comes with Node.js)
- 6.Vite Version v5.1.6
- 7.Supported Browsers: Latest versions of Chrome, Firefox, Edge, or Safari.

A.1.3 Download

The Jeecode is hosted on GitHub. Follow the steps below to download the source code:

1. Visit the Jeecode GitHub repository at https://github.com/jayesh-unde/Major-Project_Personalized-Adaptive-Recommendation-System-for-JEE
2. Click on the “code” button to open the download menu.
3. Select “download ZIP” to download the source code as zip file.
4. Once the download is complete extract the contents of the zip file to a directory on your machine. Alternatively, you can use Git to clone the repository to your machine. To do so follow these steps:

(a) Clone the Repository

Begin by cloning the project repository to your local system. This step fetches the complete project files, including the source code for both the backend and frontend components. Run the following command:

```
git clone
```

```
https://github.com/jayesh-unde/Major-Project\_Personalized-Adaptive-Recommendation-System-for-JEE.git
```

(b) After cloning, navigate to the project directory using:

```
cd Major-Project_Personalized-Adaptive-Recommendation-System-for-JEE
```

Note: Make sure to install the necessary software and code editors before proceeding with installation steps:

A.1.4 Installation

(a) Backend Dependencies:

The backend server is responsible for handling requests, processing data, and communicating with any databases or APIs.

Navigate to the backend folder, which contains the server-side code, and run the following command:

- `cd backend`
- `npm install`

Once the dependencies are downloaded, run the command given below to start the backend server:

- `cd backend`
- `node server.js`

The backend server is responsible for handling requests, processing data, and communicating with any databases or APIs. Start the server by running:

(b) Frontend Dependencies:

The frontend development server provides the user interface for the project and is typically run in development mode for testing and debugging.

Move to the frontend folder, which contains the client-side code, and install its dependencies using:

- `cd ../frontend`
- `npm install`

Once the dependencies are downloaded, run the command given below to start the frontend server:

- `cd frontend`
- `npm run dev`

A.1.5 Troubleshooting

Here are some troubleshooting steps for Jeecode web application:

Issue : npm command not recognized or Node.js version incompatible.

Solution : Verify Node.js installation using `node -v` and `npm -v` or Install the latest Node.js version from Node.js Official Website also Ensure the PATH environment variable includes the Node.js installation directory.

Issue: Vite server doesn't start, or errors like Vite command not found.

Solution: Ensure Vite is installed locally and run `npm install Vite` Confirm that Vite is listed in devDependencies in `package.json`. Check for port conflicts if the server fails to start. Use `Vite --port <custom-port>` to specify a different port or Delete `.vite` cache folder and restart the server.

REFERENCES

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LIST OF ABBREVIATIONS

API - Application Programming Interface

CF – Collaborative Filtering

CNN – Convolutional Neural Networks

CTR – Click Through Rate

GNN – Graph Neural Network

GRU – Gated Recurrent Unit

JDK – Java Development Kit

JEE – Joint Entrance Examination

JWT - JSON Web Token

LSTM – Long Short-Term Module

MAP – Mean Average Precision

ML – Machine Learning

NDCG – Normalised Discounted Cumulative Gain

NLP – Natural Language Processing

ReLU - Rectified Linear Unit

RNN – Recurrent Neural Network

UI – User Interface

UML – Unified Modelling Language